LOWER SECONDARY SCIENCE · SEC 1 & SEC 2

Lower Sec Science Cheat Sheet

Secondary 1 (Books 1A and 1B, Chapters 1–8) and Secondary 2 (Books 2A and 2B, Chapters 9–16), one card per sub-chapter, with key facts, tables to memorise, worked examples and the mistake that loses the most marks. Chapters in the EOY exam scope are marked EOY.

Secondary 1 Chapters 1–8

Book 1A Scientific Endeavour · Diversity

Ch 1 Scientific endeavour The scientific endeavour · 5 topics

What is science? 1.1

Science is the systematic study of the natural and physical world through observation and experiment, to explain how things work.

BRANCH	STUDIES
Biology	living things
Chemistry	substances and how they change
Physics	energy, forces and matter

Science vs technology: science finds out *why* and *how* things happen. Technology *applies* scientific knowledge to make useful products and solve problems (e.g. science of electricity → phones, MRT).

What is the nature of scientific knowledge? 1.2

- **Based on evidence:** ideas must be supported by observations and data that others can check
- **Tentative:** it can change when new evidence appears (e.g. the Earth-centred model of the universe was replaced by the Sun-centred model. Models of the atom changed many times)
- **Tested and shared:** scientists repeat experiments and review each other's work before accepting results
- A **theory** explains why something happens. A **law** describes what always happens

Watch out: "tentative" doesn't mean science is a guess. It means scientists change their ideas when better evidence appears.

How do we practise science? 1.3

Observe → Ask a question → Hypothesis → Plan a fair test → Collect data → Analyse → Conclude → Communicate

VARIABLE	MEANING	EXAMPLE: DOES TEMPERATURE AFFECT HOW FAST SUGAR DISSOLVES?
Independent	what you change	water temperature
Dependent	what you measure	time for sugar to dissolve
Controlled (constant)	what you keep the same for a fair test	mass of sugar, volume of water, stirring

Good practice: repeat readings and take an average, read measuring cylinders at eye level at the bottom of the meniscus, use SI units (m, kg, s, °C).

HAZARD SYMBOL	MEANING
Flame	flammable: catches fire easily
Skull and crossbones	toxic: poisonous
Liquid eating a hand and metal	corrosive: destroys skin and materials
Exclamation mark	harmful or irritant

Exploding bomb	explosive

Bunsen burner: air hole closed gives a yellow (luminous), sooty, cooler flame. Air hole open gives a blue (non-luminous), hotter flame used for heating.

Watch out: change only one variable at a time, or you can't tell which change caused the result.

What influences the way we practise science? 1.4

- **Values and attitudes:** integrity (report results honestly), objectivity (no bias), open-mindedness, curiosity, perseverance
- **Ethics:** is it right? e.g. testing on animals, using human data, genetic modification
- **Society and culture:** what people need or accept affects what is researched
- **Funding and resources:** governments and companies decide what research to pay for
- **Technology:** new instruments (microscopes, telescopes, computers) open up new discoveries
- **Safety and environment:** research must not harm people or nature

Why do we learn science? 1.5

- To understand the world around us and how our bodies work
- To make informed decisions (e.g. health, food, energy use, reading product labels)
- To solve problems in society (e.g. NEWater, vaccines, clean energy, climate change)
- To build skills: observing, questioning, reasoning, working with evidence
- For future careers in science, engineering, medicine and technology

Ch 2 Physical properties EOY

Exploring diversity of matter by its physical properties · 5 topics

Why is it important to know the physical properties of matter? 2.1 EOY

Matter is anything that has mass and takes up space. **Physical properties** can be observed or measured without changing what the substance is.

Knowing them lets us **choose the right material for a job**:

OBJECT	MATERIAL	PROPERTY USED
Cooking pot	aluminium or steel	good conductor of heat, high melting point
Pot handle	plastic or wood	poor conductor of heat
Electrical wire	copper, covered in plastic	copper conducts electricity, plastic insulates
Window	glass	transparent, hard, waterproof
Raincoat	plastic or nylon	waterproof, flexible

Watch out: when asked why a material is used, name the property and link it to the job ("copper is used because it is a good conductor of electricity, so current flows easily").

What are some common physical properties of matter? 2.2 EOY

PROPERTY	MEANING
Hardness	resists scratching (diamond is the hardest natural material)
Strength	can take a large force without breaking
Flexibility	bends without breaking
Elasticity	returns to its original shape after stretching
Malleability	can be hammered into thin sheets
Ductility	can be drawn into wires
Conductivity	lets heat or electricity pass through
Melting and boiling point	temperature at which a solid melts, or a liquid boils
Transparency	transparent (see through clearly), translucent (some light), opaque (no light)
Solubility	how well it dissolves in a liquid
Density	mass per unit volume (see 2.3)
Magnetism	attracted by a magnet (iron, steel, nickel, cobalt)

METALS	NON-METALS
shiny, good conductors of heat and electricity, malleable and ductile, usually high melting point, sonorous (ring when hit)	dull, poor conductors (except graphite), brittle when solid, usually low melting point

Watch out: mercury is a metal that is liquid at room temperature, and graphite is a non-metal that conducts electricity. These are common MCQ traps.

How do we investigate the density of objects? 2.3 EOY

$$\text{Density} = \frac{\text{mass}}{\text{volume}} \qquad \rho = \frac{m}{V} \qquad \text{units: g/cm}^3 \text{ or kg/m}^3$$

OBJECT	HOW TO FIND VOLUME
Regular solid (cuboid)	measure with a ruler: length × breadth × height
Irregular solid	**displacement**: lower it into a measuring cylinder of water. Volume = final reading − initial reading. Or use a displacement can and collect the overflow
Liquid	measuring cylinder, read at the bottom of the meniscus

Mass is measured with an electronic balance. 1 cm^3 = 1 ml. Density of water = 1 g/cm^3.

EXAMPLE
A stone has a mass of 60 g. Water rises from 50 cm^3 to 70 cm^3. Volume = 20 cm^3. Density = 60 ÷ 20 = 3 g/cm^3.

EXAMPLE
A block 2 cm × 3 cm × 5 cm has a mass of 240 g. Volume = 30 cm^3. Density = 240 ÷ 30 = 8 g/cm^3.

Watch out: density doesn't change with size. Cutting a block in half halves the mass and volume, so density stays the same.

What are the effects of different densities?

2.4 EOY

- An object **floats** if it is **less dense** than the liquid. It **sinks** if it is **denser**
- Liquids that don't mix form **layers**: the densest at the bottom (e.g. oil floats on water)
- Ice (0.92 g/cm^3) floats on water (1 g/cm^3)
- A steel ship floats because its hull holds a lot of air, so the **average density** of the whole ship is less than water
- A hot-air balloon rises because hot air is less dense than the cooler air around it
- Objects float more easily in seawater because salt water is denser than fresh water

EXAMPLE
Wood (0.6 g/cm^3) floats in water. Iron (7.9 g/cm^3) sinks in water, but floats in mercury (13.6 g/cm^3).

Watch out: heavy objects don't always sink. Compare densities, not masses.

How can understanding physical properties affect our choices for sustainable living?

2.5 EOY

Choose materials that do the job **and** harm the environment less.

CHOICE	WHY IT IS MORE SUSTAINABLE
Reusable bottles and bags (durable, strong)	used many times, so less waste
Aluminium cans, glass bottles	can be recycled many times
Biodegradable packaging (paper, plant-based)	broken down by decomposers
Good insulation in buildings	less energy needed for cooling
Lightweight (low density) materials in vehicles	less fuel needed

Most plastics are durable and cheap but **non-biodegradable**, so they stay in landfills and oceans for hundreds of years. Singapore's only landfill, Semakau, is expected to fill up, so reducing waste matters.

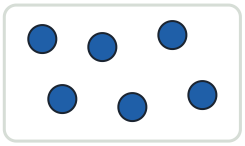
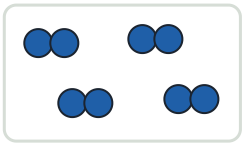
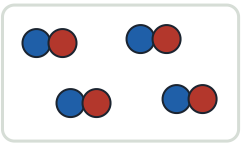
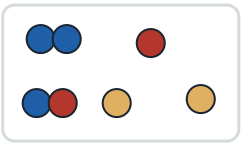
Reduce → Reuse → Recycle (the 3Rs)

Ch 3 Chemical composition EOY

Exploring diversity of matter by its chemical composition · 5 topics

Elements, compounds and mixtures compared

3.2–3.4 EOY

	ELEMENT	COMPOUND	MIXTURE
Definition	a pure substance that cannot be broken down into simpler substances by chemical means	a pure substance of two or more elements **chemically combined** in a fixed ratio	two or more substances (elements and/or compounds) **not chemically combined**
Particle diagram	  atoms (e.g. Fe) or molecules of the same atoms (e.g. O_2)	 each molecule has atoms of different elements joined (e.g. CO)	 different particles mixe not joined
Types of particle	only one kind of atom	one kind of molecule (or fixed ratio of ions)	two or more kinds of particle
Pure substance?	yes	yes	no
Composition	fixed (one element)	fixed ratio by mass, shown by a formula (water is always H_2O)	any proportion (more or l sugar in tea)
Properties	its own characteristic properties	**different** from the elements it is made of (sodium + chlorine → table salt)	each part **keeps** its own properties (iron in iron + sulfur is still magnetic)
Energy change when formed	not applicable	usually heat or light given out or taken in (a chemical reaction)	little or none (just mixing
Separated or broken down by	cannot be broken down chemically	**chemical** methods only (e.g. electrolysis of water, heating to decompose)	**physical** methods: filtration, evaporation, distillation, chromatography, magne
Melting and boiling points	fixed, sharp	fixed, sharp	melts or boils over a **ran** of temperatures
Represented by	a symbol: Fe, O, Na (or O_2 for oxygen gas)	a chemical formula: H_2O, CO_2, NaCl	no single formula: list the parts (e.g. "sand and water")
Examples	iron, copper, oxygen, hydrogen, carbon, sulfur	water, carbon dioxide, sodium chloride, glucose, calcium carbonate	air, seawater, soil, steel (alloy), milk, crude oil, orange juice

Watch out: a compound is not a mixture of elements. In a compound the elements are chemically joined and have new properties. In a mixture nothing is joined and each part keeps its properties.

How is understanding chemical composition applicable to our daily lives?

3.1 EOY

Chemical composition is what a substance is made of. Knowing it helps us:

- use substances safely (e.g. never mix bleach with other cleaners, as harmful gases can form)
- read food labels and choose healthier food (sugar, salt, fat content)
- understand medicines and their doses
- choose materials for products (e.g. alloys for coins, stainless steel for cutlery)
- detect harmful substances (e.g. carbon monoxide from burning fuel in a closed room)

What are the basic building blocks of matter and their chemically combined derivatives?

3.2 EOY

- An **element** is a pure substance that cannot be broken down into simpler substances by chemical means. It is made of only one type of atom
- There are 118 known elements, arranged in the **Periodic Table**
- Elements combine chemically to form **compounds** (their "chemically combined derivatives")

ELEMENT	SYMBOL	ELEMENT	SYMBOL
Hydrogen	H	Sodium	Na
Carbon	C	Magnesium	Mg
Nitrogen	N	Chlorine	Cl
Oxygen	O	Iron	Fe
Helium	He	Copper	Cu
Sulfur	S	Calcium	Ca

Symbols: first letter capital, second letter small (Co = cobalt, but CO = carbon monoxide, a compound).

Elements are grouped as **metals** (left and centre of the Periodic Table), **non-metals** (right) and **metalloids** (in between, e.g. silicon).

What matter comprises elements and/or compounds that are not chemically combined?

3.3 EOY

	COMPOUND	MIXTURE
Made of	two or more elements **chemically combined**	two or more substances **not chemically combined**
Proportions	fixed (water is always H_2O)	can vary (more or less sugar in tea)
Properties	different from its elements	each part keeps its own properties
Separated by	chemical reactions only	physical methods (filtering, distillation)
Energy change when formed	usually yes (heat or light)	usually little or none
Examples	water, carbon dioxide, sodium chloride (table salt)	air, seawater, alloys, soil, orange juice

> **EXAMPLE**
> Sodium (a reactive metal) and chlorine (a poisonous gas) combine to form sodium chloride, which is safe to eat. A compound's properties differ from its elements.

> **IRON + SULFUR**
> Mixed: a magnet still pulls out the iron. Heated: they react to form iron sulfide, a new compound that is not magnetic.

Types of mixture: solution (solute dissolved in a solvent, e.g. salt water, clear), **suspension** (insoluble particles spread in a liquid, e.g. muddy water, settles on standing), **alloy** (metal mixed with other elements, e.g. steel, brass).

Watch out: air is a mixture, not a compound. Its gases are not chemically combined and the proportions vary.

How can we distinguish between elements, compounds and mixtures?

3.4 EOY

See the full comparison table at the top of this chapter: Elements, compounds and mixtures compared.

- **Purity test:** a pure substance melts and boils at a fixed temperature (pure water boils at 100 °C). Impurities lower the melting point and raise the boiling point, and make them happen over a range
- **Chromatography:** a pure substance gives one spot. A mixture of dyes gives several spots

Watch out: in particle diagrams, joined circles of the same colour are an element (e.g. O_2). Joined circles of different colours are a compound. Unjoined different particles are a mixture.

Ch 4 Separation techniques EOY

Exploring diversity of matter using separation techniques · 5 topics

What is the purpose of separation techniques? 4.1 EOY

- **Obtain useful substances:** salt from seawater, sugar from sugar cane, metals from ores
- **Remove impurities or harmful substances:** clean drinking water, pure medicines
- **Identify what is in a mixture:** testing food colourings, forensic analysis of inks
- **Recycle and reduce waste:** separating metals and plastics in recycling plants

How do we choose the appropriate separation technique(s)? 4.2 EOY

Pick the technique based on the **property that differs** between the parts of the mixture.

MIXTURE	TECHNIQUE	PROPERTY USED	MORE EXAMPLES
Magnetic and non-magnetic solids (e.g. iron + sand)	**magnetic attraction**	magnetism	iron filings and sulfur powder · steel cans picked out from aluminium cans at recycling plants · iron nails mixed with sawdust
Solids of different sizes (e.g. rice + sand)	**sieving**	particle size	sifting lumps out of flour · separating gravel from sand on building sites · removing small stones from rice or beans
Insoluble solid + liquid (e.g. sand + water)	**filtration**: solid left on the filter paper is the **residue**, liquid passing through is the **filtrate**	solubility, particle size	chalk (calcium carbonate) and water · tea leaves from tea, coffee grounds from coffee · muddy water through sand filters at water works
Soluble solid from a solution (e.g. salt from salt water)	**evaporation to dryness**	boiling point	sea salt made in salt pans by the Sun · getting sodium chloride crystals back from salt solution in the lab
Soluble solid that breaks down on strong heating (e.g. copper(II) sulfate solution)	**crystallisation**: heat to a saturated solution, then cool so crystals form	solubility falls on cooling	sugar crystals from sugar-cane juice · alum crystals · blue copper(II) sulfate crystals from its solution
Pure liquid from a solution (e.g. water from seawater)	**simple distillation**: boil, then condense the vapour. The liquid collected is the **distillate**	boiling point	pure water from ink or dye solution · distilled water for science labs and car batteries · (Singapore's desalination plants use reverse osmosis instead: see Desalinated water in Singapore)
Liquids that mix, with different boiling points (e.g. ethanol + water)	**fractional distillation**	boiling point	crude oil separated into petrol, diesel, kerosene and bitumen · liquid air separated into nitrogen and oxygen · ethanol from fermented fruit juice
Liquids that don't mix (e.g. oil + water)	**separating funnel**	density, immiscibility	cooking oil and vinegar · petrol and water · removing an oil layer from water after a spill test in the lab
Dyes in ink or food colouring (e.g. black ink)	**paper chromatography**	solubility in the solvent	colourings in sweets (e.g. coated chocolates) · pigments in a leaf, such as chlorophyll · forensic tests to match the ink on a forged note

Chromatography tips: draw the start line in **pencil** (ink would run), keep it **above** the solvent level, and put a lid on to stop evaporation.

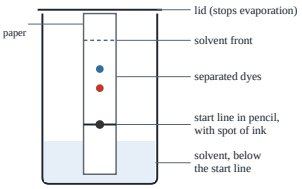
Distillation tips: thermometer bulb at the side arm, cold water enters the condenser at the **bottom**, add boiling chips for smooth boiling.

Watch out: evaporation gives you the solid and loses the water. Distillation gives you the water (and leaves the solid behind).

Paper chromatography

4.2 EOY

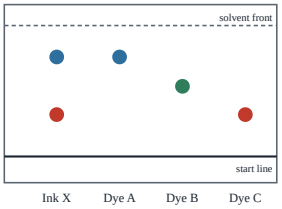
Chromatography separates a mixture of **dissolved substances**, usually dyes or colourings, by how well each one dissolves in a **solvent**. Dyes that are more soluble are carried **further** up the paper.



Method

1. Draw a **start line in pencil** about 1–2 cm from the bottom of the chromatography paper
2. Put a small, concentrated spot of the mixture (e.g. black ink) on the line
3. Stand the paper in a beaker with a little solvent (water or ethanol). The solvent must be **below** the start line. Cover the beaker
4. The solvent moves up the paper and carries the dyes with it. More soluble dyes travel further, so the mixture spreads into separate spots
5. Remove the paper before the solvent reaches the top, mark the **solvent front** and let it dry. The result is a **chromatogram**

Reading a chromatogram



WHAT YOU SEE	WHAT IT MEANS
One spot	a **pure** substance
Two or more spots	a **mixture** of dyes
Spot at the same height as a known dye (same solvent, same paper)	that dye is present
Spot stays on the start line	the substance is insoluble in this solvent

> **EXAMPLE**
> In the chromatogram above, ink X has two spots, so it is a mixture. They are at the same heights as dye A and dye C, so X contains A and C. It does not contain B.

EXAM QUESTION	ANSWER
Why draw the start line in pencil, not pen?	pen ink would dissolve in the solvent and run, mixing with the results. Pencil (graphite) does not dissolve
Why must the start line be above the solvent?	otherwise the dyes dissolve into the solvent in the beaker instead of moving up the paper
Why cover the beaker?	to stop the solvent evaporating, so it keeps moving up the paper
Why use a small, concentrated spot?	so the spots stay small and separate clearly instead of smudging into each other

Uses: checking food colourings in sweets and drinks, testing whether a substance is pure, forensic matching of inks and dyes, separating pigments in leaves (e.g. chlorophyll).

Watch out: chromatography only compares dyes fairly when the same solvent and paper are used. And "more soluble moves further" means more soluble in the solvent used.

How can we apply separation techniques to obtain potable water in Singapore?

4.3 EOY

Potable water is water that is safe to drink. Singapore gets it from the **Four National Taps**:

TAP	HOW IT WORKS
Local catchment	rainwater collected in reservoirs and drains, then treated
Imported water	water piped from Johor, Malaysia

NEWater	used water treated by microfiltration, reverse osmosis and ultraviolet (UV) disinfection
Desalination	seawater turned into fresh water, mainly by reverse osmosis

Conventional water treatment:

Screening → Coagulation (chemicals clump small particles) → Sedimentation (clumps settle) → Filtration (sand filters) → Chlorination (kills germs) → Storage

- **Microfiltration:** removes bacteria and fine particles
- **Reverse osmosis:** pressure pushes water through a partially permeable membrane, leaving salts, viruses and other dissolved substances behind
- **UV disinfection:** kills any remaining microorganisms

More detail: Desalinated water in Singapore.

Watch out: name each step and what it removes. Desalination and reverse osmosis use a lot of energy, so they are costly.

Desalinated water in Singapore

4.3 EOY

Desalination removes dissolved salts from **seawater** to make it safe to drink (potable). It is one of Singapore's **Four National Taps**, alongside local catchment water, imported water and NEWater. Because the sea never runs out and rain is not needed, it is **weather-resistant**.

Seawater intake → Pre-treatment (screens, flotation, filtration) → **Reverse osmosis** → Post-treatment (minerals added, disinfection) → Drinking water

STEP	WHAT HAPPENS	SEPARATION IDEA
1. Intake and screening	seawater is drawn in through screens that stop fish, seaweed and rubbish	like **sieving**: separates by size
2. Pre-treatment	fine particles are removed, e.g. by dissolved air flotation (bubbles carry particles to the top) and ultrafiltration membranes	like **filtration**: removes insoluble particles
3. Reverse osmosis (RO)	high pressure pushes water through a **partially permeable membrane**. Water passes through; dissolved salts and other impurities are left behind as **brine**	separates water from **dissolved** substances, like distillation does, but without boiling
4. Post-treatment	minerals are added back and the pH adjusted, then the water is disinfected (e.g. with chlorine)	makes the water safe and pleasant to drink

SINGAPORE'S DESALINATION PLANTS	OPENED
SingSpring Desalination Plant, Tuas	2005 (the first)
Tuaspring Desalination Plant, Tuas	2013
Tuas Desalination Plant	2018 (first owned by PUB)
Keppel Marina East Desalination Plant	2020 (can treat seawater *or* reservoir water, depending on the weather)
Jurong Island Desalination Plant	2022 (about 7% of daily water demand)

Desalinated water already meets a large share of Singapore's needs (roughly a quarter), and PUB plans for desalination and NEWater together to meet most of the demand by 2060.

Why reverse osmosis, not distillation? Distillation boils seawater and condenses the steam, which needs a lot of heat energy. Reverse osmosis only needs pumps to create pressure, so it uses less energy and costs less.

ADVANTAGES	DISADVANTAGES
seawater is unlimited, so it doesn't depend on rainfall or other countries	uses the most energy of the four taps, so it is expensive
reliable during dry spells and droughts	burning fuel for that energy releases carbon dioxide
plants take up little land	brine (very salty waste water) must be returned to the sea carefully to protect marine life

EXAM QUESTION	ANSWER
Name the process that removes salt from seawater in Singapore.	reverse osmosis
Explain how reverse osmosis removes salt.	high pressure forces water molecules through a partially permeable membrane. The membrane lets water through but not dissolved salts, which stay behind in the brine
Why is seawater filtered before reverse osmosis?	to remove sand, mud and particles that would block or damage the membranes
Why can't filtration alone make seawater drinkable?	salt is **dissolved**, so its particles pass through filter paper. Filtration only removes insoluble solids
Give one reason Singapore invests in desalination.	it does not depend on rainfall, so it secures water supply even during droughts

Watch out: NEWater and desalination are different taps. NEWater treats used water from homes and industries; desalination treats seawater. Both use reverse osmosis.

Book 1B Models

Ch 5 Light Ray model of light · 5 topics

How do we represent light? 5.1

- Light travels in **straight lines**. We draw it as a **ray**: a straight line with an arrow showing direction. A **beam** is many rays
- **Luminous** objects give out their own light (Sun, lamp, fire). **Non-luminous** objects are seen because they reflect light into our eyes (Moon, book)
- **Shadows** form when an opaque object blocks light, because light cannot bend around it

MATERIAL	LIGHT	EXAMPLE
Transparent	passes through almost completely, clear image	clear glass, water
Translucent	some passes through, blurred image	frosted glass, tracing paper
Opaque	none passes through	wood, metal

Watch out: we see objects because light enters our eyes. Arrows go from the object to the eye, never from the eye.

What is reflection and what are its applications? 5.2

Reflection is the bouncing of light off a surface.

Law of reflection: angle of incidence = angle of reflection $(i = r)$

- Both angles are measured from the **normal**: a line at 90° to the surface
- The incident ray, reflected ray and normal all lie in the same plane

Image in a plane mirror: same size as the object, upright, **laterally inverted** (left and right swapped), as far behind the mirror as the object is in front, and **virtual** (can't be caught on a screen).

APPLICATION	HOW IT USES REFLECTION
Periscope	two plane mirrors at 45° let you see over obstacles
Rear-view and side mirrors	let drivers see behind them
Convex mirrors at road bends and in shops	give a wider field of view
"AMBULANCE" written backwards	reads correctly in a driver's mirror (lateral inversion)

> **EXAMPLE**
> A ray hits a mirror at 30° to the surface. Angle of incidence = 90° − 30° = 60°, so angle of reflection = 60°.

Watch out: always measure angles from the normal, not from the mirror surface.

What happens when the reflecting surface is rough?

5.3

	REGULAR REFLECTION	DIFFUSE (IRREGULAR) REFLECTION
Surface	smooth (mirror, still water)	rough (paper, wall, clothing)
Reflected rays	parallel rays stay parallel	scattered in many directions
Image	clear image formed	no clear image

- The law of reflection is still obeyed at every point. The normals point in different directions because the surface is uneven, so the rays scatter
- Diffuse reflection lets us see objects from any angle, and makes paper easy to read without glare

Watch out: diffuse reflection does not break the law of reflection.

What happens when light passes through different mediums?

5.4

Marked optional for G2 Science in the textbook.

Refraction is the bending of light when it passes from one transparent medium to another, because its **speed changes**.

LIGHT TRAVELS…	SPEED	BENDS
into a denser medium (air → water or glass)	slows down	**towards** the normal
into a less dense medium (glass → air)	speeds up	**away from** the normal
along the normal (at 90° to the surface)	changes	does not bend

Effects: a swimming pool looks shallower than it is, a pencil in water looks bent, lenses in spectacles and cameras focus light.

Dispersion: a prism splits white light into the colours of the spectrum because each colour bends by a different amount:

Red → Orange → Yellow → Green → Blue → Indigo → Violet (violet bends most)

What is the impact of applications of radiation on society and the environment?

5.5

Visible light is one part of the **electromagnetic spectrum**:

Radio waves → Microwaves → Infrared → Visible light → Ultraviolet → X-rays → Gamma rays

RADIATION	USES	RISKS
Radio waves	radio, TV, communication	generally low risk
Microwaves	microwave ovens, mobile phones, satellites	heating of body tissue at high power
Infrared	remote controls, thermal cameras, heaters	burns from strong sources
Ultraviolet (UV)	sterilising water and equipment, detecting fake notes	sunburn, skin cancer, eye damage. Use sunscreen and sunglasses
X-rays	medical imaging of bones, airport security	damage to cells, so exposure is limited and staff wear lead aprons
Gamma rays	cancer treatment, sterilising food and equipment	can damage cells and cause cancer

Light pollution: too much artificial light at night disturbs animals (e.g. birds and turtles) and human sleep.

Ch 6 Cells Model of cells: the basic units of life · 3 topics

How do advances in technology help in the understanding of cells? 6.1

- Cells are too small to see with the naked eye. The **microscope** made them visible
- **Robert Hooke** (1665) saw tiny boxes in cork and named them "cells"
- **Cell theory:** all living things are made of cells, the cell is the basic unit of life, and new cells come from existing cells
- **Electron microscopes** magnify far more than light microscopes, showing tiny parts such as mitochondria in detail

Total magnification = eyepiece lens × objective lens (e.g. 10× × 40× = 400×)

MICROSCOPE PART	JOB
Eyepiece	lens you look through
Objective lenses	main magnifying lenses (low, medium, high power)
Stage and clips	hold the slide
Coarse and fine focus knobs	bring the image into focus (use fine focus at high power)
Light source or mirror	shines light through the specimen

Watch out: always start with the lowest power objective lens, then switch to higher power.

How can we use models to represent typical plant and animal cells? 6.2

PART	JOB	PLANT	ANIMAL
Cell membrane	controls what enters and leaves the cell	✓	✓
Cytoplasm	jelly where chemical reactions take place	✓	✓
Nucleus	controls cell activities, contains genetic material (DNA)	✓	✓
Mitochondria	respiration takes place here, releasing energy	✓	✓
Cell wall	made of cellulose, gives the cell a fixed shape and support	✓	✗
Chloroplasts	contain chlorophyll, absorb light for photosynthesis	✓ (green parts only)	✗
Vacuole	stores cell sap and keeps the cell firm	one large central	small, temporary

SPECIALISED CELL	ADAPTATION
Red blood cell	no nucleus and biconcave shape, so more room for haemoglobin to carry oxygen
Root hair cell	long extension gives a large surface area to absorb water and minerals
Nerve cell	long fibres carry messages around the body
Sperm cell	tail to swim, many mitochondria for energy
Guard cells	open and close stomata to control gas exchange and water loss

Watch out: root cells have no chloroplasts because they get no light. "All plant cells have chloroplasts" is false.

How does an organism function efficiently?

6.3

Cell → Tissue → Organ → Organ system → Organism

LEVEL	MEANING	HUMAN EXAMPLE	PLANT EXAMPLE
Cell	basic unit of life	muscle cell	palisade cell
Tissue	group of similar cells doing one job	muscle tissue	xylem tissue
Organ	different tissues working together	heart, stomach	leaf, root
Organ system	organs working together	circulatory, digestive system	shoot system
Organism	a complete living thing	human	plant

Division of labour: different cells, tissues and organs are specialised for different jobs, so the organism works more efficiently than if every cell did everything.

Ch 7 Particulate nature EOY

Model of matter: the particulate nature of matter · 4 topics

What is the particulate nature of matter?

7.1 EOY

Particulate model of matter:

- All matter is made of tiny, discrete **particles** (atoms, molecules or ions)
- There are **spaces** between the particles
- The particles are in **constant random motion**
- There are forces of attraction between the particles

Evidence:

- **Diffusion:** the smell of perfume spreads across a room
- **Dissolving:** sugar seems to disappear in water as its particles spread between water particles
- Mixing 50 cm^3 of alcohol with 50 cm^3 of water gives **less than** 100 cm^3, because the smaller particles fit into the spaces between the larger ones
- **Brownian motion:** smoke or pollen grains jiggle randomly because invisible air or water particles hit them

How can we explain the properties of different states of matter using the particulate nature of matter?

7.2 EOY

	SOLID	LIQUID	GAS
Arrangement	closely packed, regular pattern	closely packed, irregular	far apart, random
Movement	vibrate about fixed positions	slide past each other	move quickly in all directions
Forces between particles	very strong	strong	very weak
Shape	fixed	takes the shape of the container	fills the container
Volume	fixed	fixed	not fixed
Compressible?	no	hardly	yes, large spaces between particles

EXAMPLE

Why can a gas be compressed but a liquid cannot? Gas particles are far apart with large spaces, so they can be pushed closer together. Liquid particles are already closely packed.

Watch out: explain properties using particles: arrangement, spacing, movement and forces. "Solids are hard" alone scores nothing.

How does the particulate nature of matter explain the behaviour of particles upon heating and cooling?

7.3 EOY

- **Heating:** particles gain energy and move or vibrate faster. They move further apart, so the substance **expands**
- **Cooling:** particles lose energy, move slower and closer together, so the substance **contracts**

CHANGE	FROM → TO	PARTICLE EXPLANATION
Melting	solid → liquid	particles gain enough energy to overcome strong forces and leave fixed positions
Boiling / evaporation	liquid → gas	particles gain enough energy to break free and move far apart
Condensation	gas → liquid	particles lose energy, slow down and come closer together
Freezing	liquid → solid	particles lose energy and settle into fixed positions
Sublimation	solid → gas directly	e.g. dry ice, iodine

- During melting and boiling the temperature stays **constant**, as energy goes into overcoming forces between particles (flat parts of a heating curve)
- **Boiling:** at a fixed temperature, throughout the liquid, with bubbles. **Evaporation:** at any temperature, only at the surface
- Evaporation is faster when it is hotter, windier, less humid, or the surface area is larger

Watch out: particles themselves do not expand or melt. They move faster and further apart.

How does the particulate nature of matter help us appreciate other phenomena?

7.4 EOY

Diffusion: the net movement of particles from a region of higher concentration to lower concentration, caused by their random motion.

- Fastest in gases, slower in liquids, almost none in solids
- Faster at higher temperatures (particles move faster)
- Lighter particles diffuse faster than heavier ones

Gas pressure: gas particles hit the walls of their container. Each collision pushes on the wall. Heating makes particles move faster and hit harder and more often, so pressure rises (e.g. tyres feel harder on a hot day, an aerosol can may burst if heated).

Other examples: wet clothes dry (evaporation), a dented ping-pong ball pops back in hot water (gas inside expands), a balloon shrinks in the fridge (gas contracts), salt dissolves faster in hot water.

Watch out: pressure comes from particles colliding with the walls, not from particles getting bigger.

Ch 8 Atoms & molecules EOY

Model of matter: atoms and molecules · 4 topics

What is the simplest unit of an element and how do we represent its structure?

8.1 EOY

An **atom** is the smallest particle of an element that keeps its chemical properties.

PARTICLE	CHARGE	RELATIVE MASS	WHERE
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	≈ 1/1840 (almost 0)	shells around the nucleus

- The **nucleus** is tiny, holds almost all the mass and is positively charged. Most of an atom is empty space
- A neutral atom has equal numbers of protons and electrons
- Electrons fill shells in order: **2, 8, 8** (first 20 elements)

How the model changed: Dalton (solid sphere) → Thomson (electrons in a positive "pudding") → Rutherford (small, dense nucleus) → Bohr (electrons in shells). This shows scientific knowledge changes with new evidence.

> **EXAMPLE**
> Sodium has 11 electrons: arrangement 2, 8, 1.

How can the number of protons in an atom be used to identify an element?

8.2 EOY

- **Proton number (atomic number):** the number of protons. **Every element has its own proton number**, so it identifies the element
- **Nucleon number (mass number):** protons + neutrons
- The Periodic Table lists elements in order of proton number

Neutrons = nucleon number − proton number

ELEMENT	PROTON NO.	NUCLEON NO.	P	N	E	ELECTRON ARRANGEMENT
Hydrogen	1	1	1	0	1	1
Carbon	6	12	6	6	6	2, 4
Oxygen	8	16	8	8	8	2, 6
Sodium	11	23	11	12	11	2, 8, 1
Chlorine	17	35	17	18	17	2, 8, 7

Isotopes: atoms of the same element with the same number of protons but different numbers of neutrons (e.g. carbon-12 and carbon-14).

Watch out: the number of neutrons can vary, and electrons can be gained or lost, but if the number of protons changes, it becomes a different element.

How do we represent the simplest units of elements and some compounds?

8.3 EOY

- A **molecule** is two or more atoms chemically joined together
- Some elements exist as molecules of the same atoms: H_2, O_2, N_2, Cl_2 (diatomic)
- Compounds are made of atoms of **different** elements joined together
- A **chemical formula** shows which elements are present and how many atoms of each

SUBSTANCE	FORMULA	ATOMS IN ONE UNIT
Oxygen gas	O_2	2 oxygen
Water	H_2O	2 hydrogen, 1 oxygen
Carbon dioxide	CO_2	1 carbon, 2 oxygen
Methane	CH_4	1 carbon, 4 hydrogen
Ammonia	NH_3	1 nitrogen, 3 hydrogen
Sodium chloride	NaCl	sodium and chlorine in a 1 : 1 ratio

A number in front counts whole molecules: $3H_2O$ means 3 water molecules, so 6 hydrogen atoms and 3 oxygen atoms.

Ions: charged particles formed when atoms lose electrons (positive ions, e.g. Na^+) or gain electrons (negative ions, e.g. Cl^-). Sodium chloride is made of ions, not molecules.

Watch out: the small number after a symbol applies only to the atom just before it. In CO_2 there is 1 carbon and 2 oxygen.

What are some applications of atomic technologies and the possible issues that can arise from them?

8.4 EOY

APPLICATION	HOW IT IS USED
Nuclear power	splitting large atoms (nuclear fission) releases a lot of energy to generate electricity with little carbon dioxide
Medicine	radiotherapy kills cancer cells. Radioactive tracers show how organs work. Gamma rays sterilise equipment
Carbon dating	the amount of carbon-14 left shows the age of ancient remains
Smoke detectors, food irradiation	small radioactive sources detect smoke. Radiation kills bacteria in food

POSSIBLE ISSUE	DETAILS
Radiation exposure	damages living cells and can cause cancer
Nuclear accidents	e.g. Chernobyl (1986) and Fukushima (2011) released radiation over large areas
Radioactive waste	stays dangerous for thousands of years and must be stored safely
Weapons	the same technology can be used to make nuclear weapons

Watch out: for "evaluate" questions, give a benefit and a risk, then a judgement with a reason.

Secondary 2 Chapters 9–16

Book 2A Interactions

Ch 9 Forces & energy Application of forces and transfer of energy · 5 topics

How do forces in nature affect us? 9.1

A **force** is a push or a pull. It is measured in **newtons (N)** with a spring balance (newton meter).

A force can:

- start or stop an object moving
- make a moving object speed up or slow down
- change the direction of motion
- change the shape or size of an object

FORCE IN NATURE	EFFECT ON US
Gravity	keeps us on the ground, makes objects fall, holds the Moon in orbit, causes tides
Friction	lets us walk without slipping, lets vehicles brake
Wind (air pushing)	moves sailboats and turns wind turbines, can damage buildings in storms
Water (moving water pushing)	erodes riverbanks and coasts, turns water turbines
Magnetism	Earth's magnetic field makes a compass point north

Watch out: a force cannot be seen. We only see its effects. Name the effect in your answer (e.g. "the ball changes direction").

What are the types of forces? 9.2

CONTACT FORCES (OBJECTS TOUCH)	NON-CONTACT FORCES (ACT AT A DISTANCE)
Friction: opposes motion between surfaces in contact	**Gravitational force:** attraction between all masses
Air resistance / drag: friction from air or water	**Magnetic force:** between magnets, or magnets and iron/steel/nickel/cobalt
Normal (reaction) force: a surface pushing back on an object	**Electrostatic force:** between charged objects. Like charges repel, unlike attract
Tension: pull in a stretched rope or string	
Upthrust: upward push from a liquid or gas	

Mass vs weight

Weight = mass × gravitational field strength $W = mg$

	MASS	WEIGHT
Meaning	amount of matter in an object	gravitational pull on the object
Unit	kilogram (kg)	newton (N)
Measured with	beam or electronic balance	spring balance
Changes on the Moon?	No	Yes, about 1/6 of Earth's

Friction: useful for walking, braking and gripping. A nuisance because it wastes energy as heat and wears parts out. **Reduce** it with lubricants (oil), ball bearings, smooth surfaces and streamlined shapes. **Increase** it with rough surfaces and tyre treads.

> **EXAMPLE**
> A 50 kg student on Earth (g = 10 N/kg): weight = 50 × 10 = 500 N.

Watch out: "My weight is 50 kg" is wrong in science. Mass is in kg, weight is in N.

What happens when two or more objects interact?

9.3

- Forces always act in **pairs**. If A pushes B, B pushes back on A with an equal force in the opposite direction (e.g. a rocket pushes gas down, the gas pushes the rocket up)
- When several forces act on one object, add them up to get the **resultant force**

FORCES	RESULTANT	EFFECT
Balanced (equal and opposite)	0 N	object stays still, or keeps moving at the same speed in a straight line
Unbalanced	not 0 N	object speeds up, slows down or changes direction

Same direction: add · Opposite directions: subtract (bigger − smaller), direction of the bigger force

> **EXAMPLE**
> A box is pushed right with 30 N and friction acts left with 10 N. Resultant = 30 − 10 = 20 N to the right, so the box speeds up to the right.

> **EXAMPLE**
> A skydiver at steady speed: weight (down) = air resistance (up). Forces are balanced, so speed stays constant.

Watch out: balanced forces do not mean the object is still. It can be moving at a steady speed. Always give the direction of a resultant force.

How is energy conserved?

9.4

Principle of conservation of energy: energy cannot be created or destroyed. It can only be converted from one form to another. The total amount stays the same.

FORM	EXAMPLE
Kinetic	moving car, flowing water
Gravitational potential	book on a high shelf, water in a dam
Elastic potential	stretched rubber band, compressed spring
Chemical potential	food, fuel, batteries
Electrical, light, sound, thermal (heat), nuclear	current in wires, lamp, speaker, hot drink, Sun's core

Energy conversions:

Torch: chemical → electrical → light + heat

Falling ball: gravitational potential → kinetic (→ heat and sound on hitting the ground)

Pendulum: gravitational potential ⇄ kinetic

Work done = force × distance moved in the direction of the force $W = F \times d$ (J)

Power = energy converted ÷ time $P = \frac{E}{t}$ (W = J/s)

In real machines some energy is always converted to **wasted** forms (usually heat and sound), so the useful output is less than the input. The energy is not lost, just not useful.

> **EXAMPLE**
> Lifting a 20 N bag up 1.5 m: $W = 20 \times 1.5 = 30$ J. Done in 3 s: $P = 30 \div 3 = 10$ W.

Watch out: "energy is used up" or "lost" is wrong. Say energy is converted to heat or sound that spreads to the surroundings.

What are the sources of energy that we use?

9.5

	RENEWABLE	NON-RENEWABLE
Meaning	replaced naturally and will not run out	limited, takes millions of years to form, will run out

Examples	solar, wind, hydroelectric, tidal, geothermal, biomass	fossil fuels (coal, oil, natural gas), nuclear fuel (uranium)

SOURCE	ADVANTAGE	DISADVANTAGE
Fossil fuels	reliable, large amounts of energy, power stations easy to control	release carbon dioxide (global warming) and air pollutants, will run out
Solar	clean, free once installed	only works in daylight, affected by cloud, needs large area
Wind	clean, no fuel cost	depends on wind, noisy, needs space
Hydroelectric	clean, reliable	dams flood land and destroy habitats
Nuclear	huge energy from little fuel, no carbon dioxide while running	radioactive waste, risk of accidents
Biomass	uses waste, renewable	burning releases carbon dioxide, uses land

Singapore: most electricity is generated from imported natural gas. With little land and weak winds, solar is the main renewable source (e.g. solar panels on HDB rooftops and floating solar farms on reservoirs).

Most energy comes from the Sun: Sun → plants (photosynthesis) → fossil fuels / food

Watch out: nuclear energy is non-renewable (uranium runs out), even though it produces little carbon dioxide.

Ch 10 Heat Transfer of heat energy and its effects · 5 topics

What does a change in temperature indicate?

10.1

	TEMPERATURE	HEAT
Meaning	how hot or cold something is. A measure of the average kinetic energy of its particles	energy transferred from a hotter to a colder region because of a temperature difference
Unit	degree Celsius (°C) or kelvin (K)	joule (J)
Measured with	thermometer, temperature sensor	calculated, not measured directly

- **Temperature rises:** the object gains heat, so its particles move faster
- **Temperature falls:** the object loses heat, so its particles move slower
- Heat flows from **hot to cold** until both reach the same temperature (**thermal equilibrium**)
- A bigger object at the same temperature has more heat energy (more particles), e.g. a bathtub of warm water holds more heat than a cup at the same temperature

Reference points: ice melts at 0 °C, water boils at 100 °C (at normal pressure), human body about 37 °C.

Watch out: heat and temperature are not the same. "The temperature of the soup is 300 J" is wrong.

What effects does heat transfer have on an object?

10.2

1. Expansion and contraction

- Heated: particles vibrate or move faster and further apart, so the object **expands**. Cooled: it **contracts**
- Gases expand most, then liquids, then solids

USE OR PROBLEM	EXPLANATION
Gaps in railway tracks and bridges	leave room to expand on hot days so they don't buckle
Overhead cables sag in hot weather	they expand and get longer
Liquid-in-glass thermometer	liquid expands up the narrow tube when heated
Bimetallic strip (fire alarm, thermostat)	two metals expand by different amounts, so the strip bends
Hot water loosens a tight metal lid	the metal lid expands more than the glass jar

2. Change of state

Solid ⇄ Liquid ⇄ Gas (melting / freezing, boiling or evaporation / condensation)

- Gaining heat: melting, boiling, evaporation, sublimation. Losing heat: freezing, condensation
- Temperature stays **constant** during melting and boiling. The heat is used to break the forces between particles
- **Boiling** happens at a fixed temperature throughout the liquid. **Evaporation** happens at any temperature, only at the surface

Watch out: particles do not expand. The spaces between them get bigger.

How does the transfer of heat energy occur?

10.3

METHOD	HOW IT WORKS	HAPPENS IN
Conduction	vibrating particles pass energy to neighbouring particles. In metals, free electrons also carry energy, so metals conduct fast	mainly solids
Convection	the heated fluid expands, becomes less dense and rises. Cooler, denser fluid sinks to take its place, forming a **convection current**	liquids and gases only
Radiation	energy travels as infrared waves. No particles needed	everywhere, including a vacuum (how the Sun's heat reaches Earth)

- **Good conductors:** metals (copper, aluminium, iron). **Insulators** (poor conductors): plastic, wood, glass, air, wool, polystyrene
- Materials that trap air (wool, feathers, foam) are good insulators because air is a poor conductor and trapped air cannot form convection currents
- Convection examples: heater near the floor, air-conditioner near the ceiling, sea and land breezes, heating water in a kettle from the bottom

> **EXAMPLE: SEA BREEZE (DAY)**
> Land heats up faster than the sea. Air above the land is heated, expands and rises. Cooler air from over the sea moves in to replace it.

Watch out: convection cannot happen in solids because their particles cannot move from place to place. Never write "heat rises". Say "hot air rises because it is less dense".

What affects the rate at which heat energy is transferred?

10.4

FACTOR	FASTER TRANSFER WHEN…
Temperature difference	the difference between object and surroundings is larger
Surface area	exposed surface area is larger (e.g. fins on a motorbike engine, spreading out hot food)
Material	the material is a good conductor (metal) rather than an insulator
Thickness	the material is thinner
Surface colour and texture (radiation)	the surface is dull and black (good absorber and emitter). Shiny and white surfaces are poor absorbers and emitters, and reflect radiation
Movement of air or liquid	air or liquid flows over the surface (e.g. fan, stirring)

> **INVESTIGATION TIP**
> To test one factor (e.g. colour), keep everything else the same: same volume of water, same starting temperature, same can size, same time. Record temperature every minute.

Watch out: in an experiment question, name the variable you change, the one you measure, and at least two you keep constant.

How can we harness heat transfer in daily life?

10.5

APPLICATION	HOW IT USES HEAT TRANSFER
Cooking pot	metal base (good conductor) heats food fast. Plastic or wooden handle (insulator) is safe to hold
Vacuum flask	vacuum stops conduction and convection. Silvered walls reflect radiation. Stopper prevents convection and evaporation
Solar water heater	black pipes or panels absorb radiation from the Sun
Buildings in Singapore	light-coloured roofs reflect radiation, ventilation and high ceilings let hot air rise and escape, greenery shades walls
Refrigerator	freezing unit at the top so cold air sinks and sets up convection currents
Double-glazed windows	trapped air or vacuum between panes reduces conduction
Cooler box, polystyrene cup	insulating material with trapped air reduces conduction
Clothing	light colours in hot weather reflect radiation. Layers trap air to keep warm in cold places

Watch out: explain each feature with the method it controls: "the vacuum prevents heat loss by conduction and convection because there are no particles".

Ch 11 Chemical changes Identifying, types, how they occur, daily life · 4 topics

How do we identify a chemical change? 11.1

	PHYSICAL CHANGE	CHEMICAL CHANGE
New substance formed?	No	**Yes**
Reversible?	usually easily reversed	usually not easily reversed
Energy change	small	often large (heat or light given out or taken in)
Examples	melting ice, dissolving sugar, cutting paper, boiling water	burning paper, rusting, cooking an egg, baking a cake, fruit ripening

Signs that a chemical change may have happened:

- colour change (e.g. iron turning reddish-brown)
- gas given off: bubbles or fizzing (effervescence)
- a **precipitate** (insoluble solid) forms when two solutions are mixed
- temperature change: gets hot or cold
- light or sound produced
- change in smell

Watch out: signs only suggest a chemical change. Boiling water also gives off bubbles but no new substance forms. The only proof is a new substance.

What are the different types of chemical changes? 11.2

TYPE	WHAT HAPPENS	WORD EQUATION	BALANCED CHEMICAL EQUATION
Combustion (burning)	a substance reacts with oxygen, giving out heat and light	methane + oxygen → carbon dioxide + water	$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$
Oxidation	a substance combines with oxygen, often slowly	iron + oxygen + water → rust (hydrated iron(III) oxide)	$4Fe(s) + 3O_2(g) + 2xH_2O(l) \rightarrow 2Fe_2O_3 \cdot xH_2O(s)$ simpler example: $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$
Thermal decomposition	heat breaks one substance into simpler ones	calcium carbonate → calcium oxide + carbon dioxide	$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$
Neutralisation	an acid reacts with an alkali or base	hydrochloric acid + sodium hydroxide → sodium chloride + water	$HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
Metal + acid	a reactive metal reacts with a dilute acid	magnesium + hydrochloric acid → magnesium chloride + hydrogen	$Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
Carbonate + acid	gives off carbon dioxide (fizzing)	calcium carbonate + hydrochloric acid → calcium chloride + water + carbon dioxide	$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$
Reaction caused by light	light energy drives the change	photosynthesis: carbon dioxide + water → glucose + oxygen	$6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(aq) + 6O_2(g)$ needs light and chlorophyll
Electrolysis	electricity breaks down a compound	water → hydrogen + oxygen	$2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$

State symbols: (s) solid, (l) liquid, (g) gas, (aq) aqueous, meaning dissolved in water. In rust, *x* means the amount of water can vary.

GAS TEST	RESULT
Hydrogen: lighted splint	"pop" sound
Oxygen: glowing splint	relights
Carbon dioxide: bubble through limewater	white precipitate (limewater turns chalky)

Watch out: rusting needs both oxygen and water. Iron in dry air or in boiled water with an oil layer does not rust.

How do chemical changes occur?

11.3

- In a chemical change, the atoms in the **reactants** are rearranged to form new **products**. Bonds between atoms are broken and new bonds are formed
- No atoms are created or destroyed, so the **total mass stays the same** (conservation of mass). In an open container, mass may seem to drop if a gas escapes
- Reactions need particles to **collide**. They happen faster with higher temperature, higher concentration, smaller pieces (larger surface area) or a catalyst

reactants → products

ENERGY CHANGE	MEANING	EXAMPLES
Exothermic	gives out heat, surroundings get warmer	combustion, neutralisation, respiration, hand warmers
Endothermic	takes in heat, surroundings get colder	thermal decomposition, photosynthesis, instant cold packs

Writing equations: word equation first, then a balanced chemical equation by counting atoms on both sides. Only change the numbers in front.

$2Mg + O_2 \rightarrow 2MgO \quad CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Watch out: never change the small numbers inside a formula when balancing. H_2O must stay H_2O.

How does knowledge of chemical reactions help us in daily life?

11.4

USE	CHEMISTRY BEHIND IT
Preventing rust	paint, oil or grease, plastic coating keep out oxygen and water. Galvanising (zinc coating) protects iron even when scratched
Preserving food	refrigeration slows reactions. Vacuum packing and canning remove oxygen. Antioxidants slow oxidation (e.g. lemon juice stops cut apples browning)
Fire safety	fire needs fuel, oxygen and heat (the **fire triangle**). Remove one to put it out: fire blanket or CO_2 extinguisher removes oxygen, water removes heat
Antacids	a base neutralises excess stomach acid
Baking	baking powder (sodium hydrogencarbonate) releases carbon dioxide when heated, making cakes rise
Fuels	combustion releases energy for cooking, transport and electricity
Treating acidic soil	farmers add lime (a base) to neutralise it

Watch out: never use water on an oil or electrical fire. Water spreads burning oil and conducts electricity.

Ch 12 Ecosystems Interactions within ecosystems · 5 topics

How do our actions impact the environment? 12.1

HUMAN ACTION	IMPACT
Deforestation (for farms, timber, buildings)	habitat loss and extinction, soil erosion, less carbon dioxide absorbed
Burning fossil fuels	more carbon dioxide → enhanced greenhouse effect → global warming, rising sea levels. Air pollution and haze
Water pollution (sewage, fertilisers, oil, waste)	fertiliser runoff causes algae to grow fast. When algae die, decomposers use up oxygen and fish die
Land pollution (plastic, litter)	animals eat or get trapped in plastic. Microplastics enter food chains
Overfishing, hunting	populations fall faster than they can recover. Food webs are disrupted
Introducing foreign (invasive) species	they compete with or eat native species

Positive actions: conservation, reforestation, recycling, using renewable energy. See 12.5.

Watch out: link each action to a specific effect on organisms ("fewer trees → fewer habitats → fewer animals"), not just "it is bad for the environment".

How does an environment influence the survival of organisms? 12.2

- **Environment:** the surroundings of an organism, made up of **biotic** (living: plants, animals, microorganisms) and **abiotic** (non-living) factors
- **Abiotic factors:** light, temperature, water, air (oxygen, carbon dioxide), soil pH and minerals, salinity (saltiness)
- **Habitat:** the place where an organism lives
- **Adaptation:** a feature that helps an organism survive in its environment

ORGANISM	ADAPTATION	HOW IT HELPS
Cactus (desert)	spines instead of leaves, thick waxy stem, long roots	reduce water loss, store water, reach water deep underground
Camel	hump stores fat, long eyelashes, wide feet	energy and water when food is scarce, keeps sand out, walks on sand
Mangrove trees (Singapore coasts)	breathing roots (pneumatophores) stick up out of mud. Prop roots	take in oxygen from air in waterlogged mud. Support in soft mud
Polar bear	thick fur and fat, white coat	insulation, camouflage in snow
Owl	large eyes, active at night	sees in low light, hunts when prey is active

Structural adaptations are body features. **Behavioural** adaptations are actions (e.g. animals sheltering in the heat of the day, migration).

Watch out: state the feature and explain how it helps survival. "The cactus has spines" alone scores half.

What types of interactions affect the stability of an ecosystem? 12.3

INTERACTION	WHO BENEFITS	EXAMPLE
Predator–prey	predator gains, prey is killed	eagle and snake, lion and zebra
Competition	both are harmed (less resource for each)	plants competing for light and water. Animals for food, mates, space
Mutualism	both benefit (+ +)	bee and flower (nectar / pollination), clownfish and sea anemone
Commensalism	one benefits, the other is unaffected (+ 0)	barnacles on a whale, orchid growing on a tree
Parasitism	parasite benefits, host is harmed (+ −)	tick or flea on a dog, tapeworm in a human

Predator–prey cycle: more prey → more food → predator numbers rise → more prey eaten → prey numbers fall → predators starve → predator numbers fall → prey recover. The predator graph peaks **after** the prey graph.

These interactions keep population sizes in check, which keeps the ecosystem **stable**.

Watch out: commensalism and mutualism are often mixed up. Ask: does the second organism gain anything? If not, it's commensalism.

What are the processes involved in maintaining a stable ecosystem?

12.4

1. Energy flow through food chains and webs

Sun → Producer → Primary consumer → Secondary consumer → Tertiary consumer

- **Producers** (green plants) make food by photosynthesis. **Consumers** eat other organisms. **Decomposers** (bacteria, fungi) break down dead matter and waste
- Arrows show the direction of **energy flow**: from the organism eaten to the eater
- Only about **10%** of energy passes to the next level. The rest is lost as heat (respiration), movement and waste. So food chains are short and there are fewer organisms at higher levels
- A **food web** is many linked food chains. More links make an ecosystem more stable

2. Cycling of nutrients: the carbon cycle

REMOVES CO_2 FROM AIR	ADDS CO_2 TO AIR
photosynthesis by plants	respiration by all living things, combustion of fuels, decomposition by decomposers

Photosynthesis: carbon dioxide + water → glucose + oxygen (light, chlorophyll)

Respiration: glucose + oxygen → carbon dioxide + water + energy

> **EXAMPLE**
> In a food web, if snakes are removed, frogs increase (fewer eaten), so insects decrease (more eaten by frogs).

Watch out: energy flows one way and is not recycled. Nutrients like carbon are recycled.

How can we help to maintain a stable ecosystem?

12.5

ACTION	HOW IT HELPS	EXAMPLE
Protect habitats	keeps food webs complete and species safe	Central Catchment and Bukit Timah Nature Reserves, Sungei Buloh Wetland Reserve
Replant and restore	creates habitats and absorbs carbon dioxide	mangrove replanting, OneMillionTrees movement
Reduce, reuse, recycle	less waste and pollution, fewer resources used	bring own bag and bottle, recycling bins
Save energy, use renewables	less carbon dioxide from burning fossil fuels	solar panels, switching off appliances
Sustainable fishing and farming	populations can recover	fishing quotas, buying certified seafood
Laws and education	stop illegal hunting, trade and dumping	bans on wildlife trade, campaigns

Watch out: say how the action helps the ecosystem, e.g. "recycling paper means fewer trees are cut down, so habitats are kept".

Book 2B Systems

Ch 13 Electrical systems Circuits, components, safety, saving energy · 5 topics

What does an electrical system comprise?

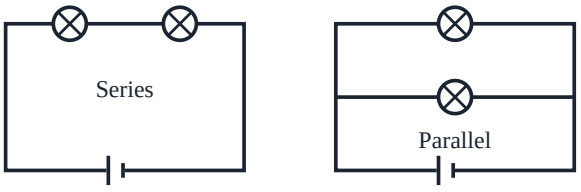
13.1

PART	JOB	EXAMPLES

Source	provides electrical energy (pushes the current)	cell, battery, mains supply, solar panel
Conductor	carries current between components	copper wires
Load	converts electrical energy to other forms	bulb, motor, buzzer, heater
Switch	opens or closes the circuit	push switch, light switch

Current only flows in a **closed circuit** (a complete loop with no gaps).

QUANTITY	MEANING	UNIT	MEASURED WITH
Current I	rate of flow of electric charge	ampere (A)	ammeter, connected in **series**
Voltage V	electrical "push" from the source, or energy used by a component	volt (V)	voltmeter, connected in **parallel**
Resistance R	how much a component opposes current	ohm (Ω)	calculated: $R = V \div I$
Power P	rate of converting electrical energy	watt (W)	calculated: $P = V \times I$ (see 13.4)



	SERIES	PARALLEL
Layout	one loop	two or more branches
Current	same everywhere	splits between branches, adds up to the total
Voltage	shared between components	same across each branch
One bulb blows	all go out	others stay on (used in homes)

Conventional current flows from the positive (+, long line) to the negative (−, short line) terminal. Electrons actually flow the other way.

Watch out: ammeter in series, voltmeter in parallel. Swapping them is a common lost mark in circuit diagrams.

How do electrical components affect the output of an electrical system?

13.2

Marked optional for G2 Science in the textbook.

$V = I \times R$ (voltage = current × resistance)

CHANGE	EFFECT ON CURRENT	EFFECT ON OUTPUT
Add more cells in series	increases	bulbs brighter, motor faster
Add more bulbs in series	decreases (more resistance)	each bulb dimmer
Add more bulbs in parallel	total current increases	each bulb as bright as one on its own (battery runs down faster)
Increase resistance (e.g. variable resistor)	decreases	bulb dimmer, motor slower

Resistance of a wire is higher when it is: longer, thinner, made of a poorer conductor (e.g. nichrome rather than copper), or hotter.

A **variable resistor (rheostat)** changes resistance to control the output, e.g. a dimmer switch or a fan speed control.

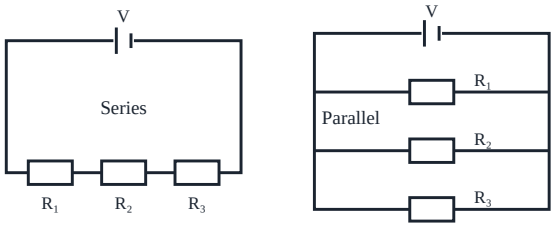
> **EXAMPLE**
> A bulb has 3 V across it and a current of 0.5 A. $R = 3 \div 0.5 = 6$ Ω.
> A 12 V supply drives current through a 4 Ω resistor. $I = 12 \div 4 = 3$ A.

Watch out: rearrange the formula before substituting, and always write the unit (A, V or Ω).

Total resistance in series and parallel circuits

13.2

How to find the **total (effective) resistance** of several resistors, and why the formulas work. Every step uses only $V = I \times R$.



Series: same current through every resistor

1. Each resistor: $V_1 = IR_1$, $V_2 = IR_2$, $V_3 = IR_3$
2. The voltages share the supply: $V = V_1 + V_2 + V_3 = I(R_1 + R_2 + R_3)$
3. For the whole circuit, $V = I \times R_{\text{total}}$, so

$$R_{\text{total}} = R_1 + R_2 + R_3$$

Parallel: same voltage across every branch

1. Each branch has the full supply voltage: $V_1 = V_2 = V_3 = V$
2. Each branch current: $I_1 = \frac{V}{R_1}$, $I_2 = \frac{V}{R_2}$, $I_3 = \frac{V}{R_3}$
3. The currents add up: $I_{\text{total}} = I_1 + I_2 + I_3$
4. $R_{\text{total}} = \frac{V}{I_{\text{total}}} = \frac{V}{\frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}}$
5. Flip both sides and divide through by V:

$$\frac{1}{R_{\text{total}}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Shortcut for two resistors in parallel: $R_{\text{total}} = \frac{R_1 \times R_2}{R_1 + R_2}$ (product over sum)

	SERIES	PARALLEL
Current	same everywhere: I	adds up: $I_{\text{total}} = I_1 + I_2 + I_3$
Voltage	adds up: $V = V_1 + V_2 + V_3$	same across each branch: V
Total resistance	$R_1 + R_2 + R_3$ (bigger than any one)	$1/R = 1/R_1 + 1/R_2 + 1/R_3$ (smaller than the smallest)

EXAMPLE 1: SERIES

2 Ω, 3 Ω and 5 Ω in series with a 12 V supply.

$R_{\text{total}} = 2 + 3 + 5 =$ **10 Ω**. $I = 12 \div 10 =$ **1.2 A**.

$V_1 = 1.2 \times 2 = 2.4$ V, $V_2 = 1.2 \times 3 = 3.6$ V, $V_3 = 1.2 \times 5 = 6.0$ V. Check: 2.4 + 3.6 + 6.0 = 12 V ✓

EXAMPLE 2: TWO IN PARALLEL

6 Ω and 3 Ω in parallel with a 6 V supply.

$1/R = 1/6 + 1/3 = 1/6 + 2/6 = 3/6$, so $R_{\text{total}} =$ **2 Ω** (shortcut: 18 ÷ 9 = 2 Ω).

$I_1 = 6 \div 6 = 1$ A, $I_2 = 6 \div 3 = 2$ A, $I_{\text{total}} =$ **3 A**. Check: 6 V ÷ 2 Ω = 3 A ✓

EXAMPLE 3: THREE IN PARALLEL

2 Ω, 3 Ω and 6 Ω in parallel.

$1/R = 1/2 + 1/3 + 1/6 = 3/6 + 2/6 + 1/6 = 6/6 = 1$, so $R_{\text{total}} =$ **1 Ω**.

Watch out: in parallel, $1/R_{\text{total}}$ is not the answer. Remember to flip it at the end ($1/R = 3/6$ means $R = 2$ Ω, not 0.5 Ω). Check: the parallel total must be smaller than the smallest resistor.

What are some benefits and risks associated with the applications of electrical systems?

13.3

Benefits: lighting, cooking and heating, refrigeration of food and medicine, communication (phones, internet), transport (MRT, electric buses), medical equipment, entertainment.

RISK	CAUSE
Electric shock	touching live wires, damaged insulation, wet hands or water near appliances
Fire	overloading a socket (too many appliances), short circuit, wires overheating
Short circuit	current takes a path with almost no resistance, so a very large current flows and wires overheat

SAFETY FEATURE	HOW IT PROTECTS
Fuse (in the live wire)	thin wire melts when current is too large, breaking the circuit
Circuit breaker	switch that trips off automatically when current is too large. Can be reset
Earth wire	if a live wire touches a metal casing, current flows safely to earth instead of through the user, and the fuse blows
Insulation / double insulation	plastic covering stops users touching live parts

WIRE IN A 3-PIN PLUG	COLOUR
Live	brown
Neutral	blue
Earth	green and yellow stripes

Watch out: the fuse and switch must be on the live wire, so the appliance is disconnected from the high voltage when they break the circuit.

How do we reduce the consumption of electrical energy?

13.4

Finding power from voltage and current

Power is how fast an appliance converts electrical energy, measured in **watts (W)**. 1 W = 1 joule per second.

Power = voltage × current $P = V \times I$

watts (W) = volts (V) × amperes (A)

TO FIND	USE
Power	$P = V \times I$
Current	$I = P \div V$
Voltage	$V = P \div I$

Steps: (1) write the formula, (2) substitute the values, (3) calculate, (4) give the unit.

> **EXAMPLE 1: FIND POWER**
> A kettle on Singapore's 230 V mains draws a current of 8 A.
> $P = V \times I$ = 230 × 8 = **1840 W** (1.84 kW).

> **EXAMPLE 2: FIND CURRENT**
> A 2300 W air-conditioner runs on 230 V.
> $I = P \div V$ = 2300 ÷ 230 = **10 A**.

> **EXAMPLE 3: TORCH BULB**
> A bulb has 3 V across it and 0.5 A through it.
> P = 3 × 0.5 = **1.5 W**.

> **EXAMPLE 4: POWER, THEN ENERGY AND COST**
> A 230 V heater draws 5 A for 2 hours at $0.30 per kWh.
> P = 230 × 5 = 1150 W = 1.15 kW.
> Energy = 1.15 kW × 2 h = 2.3 kWh.
> Cost = 2.3 × $0.30 = **$0.69**.

Choosing a fuse: work out the normal current with $I = P \div V$, then pick the fuse rated just above it. A 1840 W kettle draws 8 A, so use a 13 A fuse, not a 3 A or 5 A fuse.

Linking with V = IR: if you know the current and resistance, first find the voltage ($V = I \times R$), then the power ($P = V \times I$).

Watch out: current must be in amperes. Convert mA first (250 mA = 0.25 A). And kW ≠ W: divide by 1000 before working out kWh.

Energy used and its cost

$$\text{Energy (kWh)} = \text{Power (kW)} \times \text{time (h)}$$

$$\text{Cost} = \text{energy used (kWh)} \times \text{price per kWh}$$

> **EXAMPLE**
> A 2000 W air-conditioner runs 5 h a day. Energy = 2 kW × 5 h = 10 kWh per day. At $0.30 per kWh, it costs $3.00 a day, or about $90 a month.

Ways to save electricity:

- Switch off lights and appliances when not in use, including at the wall socket (standby still uses power)
- Use LED bulbs: they convert more energy to light and less to heat than older bulbs
- Buy appliances with more ticks on the **Energy Label**: more ticks means more energy efficient
- Set the air-conditioner to 25 °C or higher, or use a fan instead
- Use natural light and ventilation, and keep fridge doors closed

Why it matters: most electricity in Singapore comes from burning natural gas, which releases carbon dioxide. Using less electricity lowers both the bill and emissions.

Watch out: convert W to kW (÷ 1000) and minutes to hours (÷ 60) before calculating kWh.

Ch 14 Digestive system Human digestive system · 4 topics

Why is a digestive system needed in humans? 14.1

Most food molecules (starch, proteins, fats) are **large and insoluble**. They cannot pass through the gut wall into the blood. **Digestion** breaks them down into **small, soluble** molecules that can be absorbed and used by cells.

TYPE	WHAT HAPPENS	EXAMPLES
Physical digestion	breaks food into smaller pieces, increasing surface area for enzymes. No new substance	chewing by teeth, churning in stomach, bile emulsifying fats
Chemical digestion	**enzymes** break large molecules into small ones	amylase breaking starch into maltose

NUTRIENT	BROKEN DOWN INTO
Starch (carbohydrate)	glucose
Protein	amino acids
Fats	fatty acids and glycerol

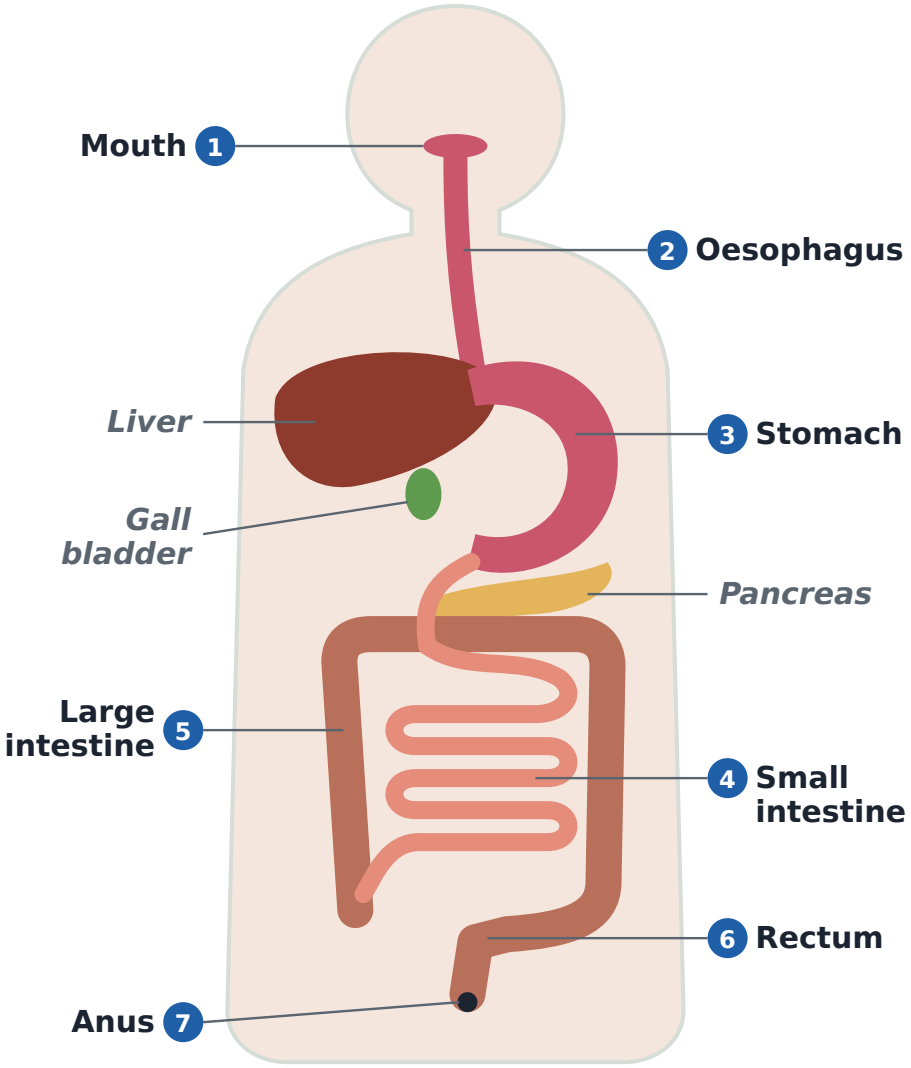
Vitamins, minerals and water are already small and are absorbed without digestion. **Fibre** is not digested but helps food move through the gut.

FOOD TEST	REAGENT	POSITIVE RESULT
Starch	iodine solution	brown → blue-black
Reducing sugar	Benedict's solution, heated	blue → brick-red precipitate
Protein	biuret solution	blue → violet
Fats	ethanol, then water	white emulsion

Watch out: the reason for digestion is size and solubility. Mention both.

How do the parts of the human digestive system work together?

14.2



1–7 = the path food takes (the alimentary canal). *Liver, gall bladder and pancreas* make or store digestive juices, but food does **not** pass through them.

Mouth → Oesophagus → Stomach → Small intestine → Large intestine → Rectum → Anus

PART	WHAT HAPPENS
Mouth	teeth chew food. Saliva moistens it and contains **amylase** (starch → maltose)
Oesophagus	**peristalsis**: waves of muscle contraction push food to the stomach
Stomach	churns food. Hydrochloric acid kills bacteria and gives the right pH for **protease** (protein digestion starts)
Liver and gall bladder	liver makes **bile**, stored in the gall bladder. Bile emulsifies fats into tiny droplets
Pancreas	releases pancreatic juice with amylase, protease and lipase into the small intestine
Small intestine	digestion is completed. Digested food is **absorbed** into the blood through **villi**
Large intestine	absorbs water and minerals. Forms faeces
Rectum, anus	stores faeces, then removes them (egestion)

ENZYME	SUBSTRATE → PRODUCT	WHERE
Amylase	starch → maltose	mouth, small intestine
Maltase	maltose → glucose	small intestine
Protease	protein → amino acids	stomach, small intestine
Lipase	fats → fatty acids + glycerol	small intestine

Villi are adapted for absorption: they give a very large surface area, have walls one cell thick, and contain many blood capillaries.

Enzymes: proteins that speed up reactions. Each works on one substance (lock and key). They work best at an optimum temperature (about 37 °C in humans) and pH, and are **denatured** at high temperatures.

Watch out: food never passes through the liver, gall bladder or pancreas; they only send juices into the gut. Also, bile is not an enzyme. It breaks fat into droplets physically, which gives lipase a larger surface area to work on.

How do the end products of digestion affect the body?

14.3

After absorption, the end products are carried in the blood to every cell and used there (**assimilation**).

END PRODUCT	USE IN THE BODY	IF IN EXCESS
Glucose	used in **respiration** to release energy	stored as glycogen in the liver and muscles, or converted to fat
Amino acids	build new proteins for growth and repair, enzymes, hormones	broken down in the liver. The waste (urea) is removed by the kidneys
Fatty acids and glycerol	make cell membranes, energy store, insulation, protect organs	stored as fat under the skin and around organs

Aerobic respiration: glucose + oxygen → carbon dioxide + water + energy

Watch out: respiration releases energy in cells. It is not the same as breathing, which moves air in and out of the lungs.

What are some issues that affect the human digestive system?

14.4

ISSUE	CAUSE	PREVENTION OR CARE
Constipation	too little fibre and water, so faeces become hard	eat vegetables, fruit, wholegrains, drink water
Diarrhoea	infection from contaminated food or water. Large intestine absorbs too little water	wash hands, eat clean food, drink fluids to avoid dehydration
Gastric ulcer, gastritis	stomach lining damaged by bacteria, too much acid, some painkillers, irregular meals	regular meals, medical treatment
Tooth decay	bacteria feed on sugar and produce acid that dissolves enamel	brush twice a day, less sugary food and drinks
Obesity	taking in more energy than the body uses	balanced diet, regular exercise

A balanced diet has the right amounts of carbohydrates, proteins, fats, vitamins, minerals, fibre and water (e.g. Singapore's "My Healthy Plate": half fruit and vegetables, a quarter wholegrains, a quarter protein).

Watch out: link the cause to the part of the gut affected, e.g. "too little water is absorbed in the large intestine, so faeces are watery".

Ch 15 Transport systems

Transport in humans and plants, diffusion and osmosis · 3 topics

What are the functions of a transport system and its parts?

15.1

A transport system carries useful substances (oxygen, food, water) to cells and removes waste (carbon dioxide, urea) from them.

Humans: heart, blood vessels and blood

Vena cava → Right atrium → Right ventricle → Pulmonary artery → **Lungs** → Pulmonary vein → Left atrium → Left ventricle → Aorta → **Body**

- Right side pumps deoxygenated blood to the lungs. Left side pumps oxygenated blood to the body
- The left ventricle has the thickest wall because it pumps blood all round the body. Valves stop blood flowing backwards

VESSEL	DIRECTION	FEATURES
Artery	away from heart	thick, muscular, elastic wall. High pressure
Vein	towards heart	thin wall, wide space inside, valves. Low pressure
Capillary	links arteries and veins	wall one cell thick so substances pass in and out

BLOOD PART	JOB
Red blood cells	carry oxygen using haemoglobin. No nucleus, biconcave shape for more space and surface area
White blood cells	fight germs (engulf them or make antibodies)
Platelets	help blood clot at wounds
Plasma	liquid that carries dissolved food, carbon dioxide, hormones and urea

Plants: xylem carries water and mineral salts up from the roots (one direction). **Phloem** carries sugars made in the leaves to all parts (up and down). **Transpiration** (water evaporating from leaves through stomata) pulls water up the xylem. It is faster when it is hot, windy, bright or dry.

Watch out: the pulmonary artery carries deoxygenated blood. Arteries are named by direction (away from the heart), not by oxygen.

What are the processes involved in the movement of substances in and out of cells?

15.2

	DIFFUSION	OSMOSIS
Definition	net movement of particles from a region of **higher** concentration to a region of **lower** concentration	net movement of **water** molecules from a dilute solution (more water) to a concentrated solution (less water) through a **partially permeable membrane**
What moves	any particles (gases, dissolved substances)	water only
Examples	oxygen from air sacs into blood in the lungs. Carbon dioxide into a leaf. Digested food into villi	root hair cells taking in water from soil. Water moving between plant cells

Diffusion is faster when: the concentration difference is larger, the temperature is higher, the surface area is larger, or the distance is shorter.

CELL PLACED IN	PLANT CELL	RED BLOOD CELL (ANIMAL)
Pure water (dilute)	gains water, becomes firm (**turgid**). Cell wall stops it bursting	gains water, swells and **bursts**
Concentrated salt or sugar solution	loses water, becomes limp (**flaccid**). Plant wilts	loses water and **shrinks**

> **EXAMPLE**
> Salted cucumber slices go soft: water leaves the cucumber cells by osmosis into the more concentrated salt solution outside.

Watch out: in osmosis, describe water moving, not salt or sugar. And always mention the partially permeable membrane.

What issues arise when parts of the human transport system do not work well?

15.3

ISSUE	WHAT HAPPENS	EFFECT
Coronary heart disease	fatty deposits build up inside the coronary arteries that supply the heart muscle, narrowing them	less oxygen reaches heart muscle, causing chest pain. A blocked artery causes a **heart attack**
High blood pressure	blood pushes too hard on artery walls	damages arteries, raises risk of heart attack and stroke
Stroke	blood supply to part of the brain is blocked or a vessel bursts	brain cells die, can cause paralysis or speech problems
Anaemia	too little haemoglobin or too few red blood cells, often from lack of iron	less oxygen carried, so tiredness and paleness
Faulty heart valves	valves don't close properly, so blood flows backwards	heart pumps less efficiently

Risk factors: smoking, diet high in saturated fat and salt, lack of exercise, obesity, stress, family history.

Prevention: do not smoke, eat less fatty and salty food, exercise regularly, keep a healthy weight, have health check-ups.

Treatment: medicines, a **stent** (small tube that holds an artery open), bypass surgery, pacemaker or heart valve replacement.

Watch out: the coronary arteries supply blood to the heart muscle itself, not the blood inside the heart chambers.

Ch 16 Reproductive system Human sexual reproductive system · 4 topics

What is the function of the human sexual reproductive system? 16.1

- **Function:** to produce offspring, so the human species continues
- **Sexual reproduction** involves two parents. A male gamete (**sperm**) fuses with a female gamete (**egg** or ovum). This is **fertilisation**
- Offspring get genetic material from both parents, so they show **variation** (they are not identical)

The system: produces gametes, produces sex hormones (testosterone in males, oestrogen and progesterone in females), allows sperm to meet the egg, and (in females) protects and nourishes the developing baby.

Puberty is when the reproductive organs mature and secondary sexual characteristics develop, controlled by sex hormones.

	SPERM	EGG
Made in	testes	ovaries
Size	very small	much larger (contains food store)
Moves?	yes, swims with a tail	no, moved along the oviduct by cilia
Number	millions produced	usually one released per month

How do the parts of each reproductive system work together? 16.2

MALE ORGAN	JOB
Testes	produce sperm and testosterone
Scrotum	holds the testes outside the body, at a slightly lower temperature for sperm production
Sperm duct	carries sperm from the testes to the urethra
Prostate gland, seminal vesicle	add fluid that nourishes sperm and helps them swim (sperm + fluid = semen)
Urethra, penis	carry semen out of the body

FEMALE ORGAN	JOB
Ovaries	release an egg about once a month. Produce oestrogen and progesterone
Oviduct (fallopian tube)	carries the egg to the uterus. **Fertilisation happens here**
Uterus	embryo implants in its thick lining and the fetus develops
Cervix	ring of muscle at the neck of the uterus
Vagina	receives sperm. Birth canal

Sperm + egg → **zygote** (in oviduct) → divides into an **embryo** → implants in uterus lining → **fetus** → birth (about 9 months)

- **Placenta:** food and oxygen pass from mother to fetus. Carbon dioxide and wastes pass from fetus to mother. Their blood does not mix
- **Umbilical cord:** connects the fetus to the placenta
- **Amniotic fluid:** cushions the fetus from bumps
- **Menstrual cycle (about 28 days):** days 1–5 menstruation (uterus lining breaks down). Lining thickens again. Ovulation (egg released) around day 14. Fertile period about days 11–17

Watch out: fertilisation happens in the oviduct. The uterus is where implantation happens.

How can the function of the human reproductive system be disrupted? 16.3

DISRUPTION	DETAILS
Sexually transmitted infections (STIs)	spread through sexual contact. Examples: HIV (attacks the immune system and can lead to AIDS), syphilis and gonorrhoea (bacterial). Some can damage reproductive organs and cause infertility
Infertility	unable to conceive, e.g. blocked oviducts, egg not released, low sperm count or sperm that swim poorly

Harmful substances in pregnancy	smoking, alcohol and drugs cross the placenta and can cause low birth weight, birth defects or miscarriage
Hormone imbalance	can stop ovulation or sperm production
Age	fertility falls as people get older

Preventing STIs: abstinence is the only certain way. Condoms reduce the risk. Some STIs can spread by sharing needles, by blood, or from mother to baby.

Watch out: antibiotics treat bacterial STIs like syphilis and gonorrhoea, but not HIV, which is a virus.

How do medical advancements impact human reproduction and survival?

16.4

ADVANCEMENT	HOW IT HELPS	ISSUE TO CONSIDER
In vitro fertilisation (IVF)	eggs are fertilised by sperm outside the body, and an embryo is placed in the uterus. Helps infertile couples	expensive, may need several attempts, raises ethical questions about unused embryos
Contraception	prevents pregnancy, so families can plan. Barrier methods (condoms) stop sperm reaching the egg. Hormonal methods (pills) stop ovulation. Surgical methods (cutting the sperm ducts or oviducts) are permanent	no method is 100% effective. Only condoms reduce STI risk
Ultrasound scans and prenatal tests	check the fetus's growth and detect problems early	results can lead to difficult decisions
Vaccines	e.g. the HPV vaccine protects against a virus that can cause cervical cancer	must be given before exposure
Medicines for STIs	antibiotics cure bacterial STIs. Antiretroviral drugs control HIV so people can live longer, healthier lives	HIV has no cure yet

Watch out: "evaluate" questions want a benefit and a limitation or ethical issue, then a judgement.

Other topics

Other topics Not in Books 1A–2B · 1 topic

Acids & alkalis SEC 2



0 7 14

strong acid neutral strong alkali

- **Litmus:** acid turns blue litmus red. Alkali turns red litmus blue
- **Universal indicator:** red/orange/yellow = acid, green = neutral (pH 7), blue/purple = alkali
- Acids release H^+ ions in water. Alkalis release OH^- ions
- An alkali is a base that dissolves in water

acid + metal → salt + hydrogen

acid + carbonate → salt + water + carbon dioxide

acid + alkali → salt + water (neutralisation)

GAS	TEST	RESULT
Hydrogen	lighted splint	"pop" sound
Carbon dioxide	bubble through limewater	white precipitate
Oxygen	glowing splint	relights

Uses of neutralisation: antacids for indigestion, lime (calcium hydroxide) for acidic soil, toothpaste against mouth acid.

> **EXAMPLE**
> hydrochloric acid + sodium hydroxide → sodium chloride + water

Watch out: copper and other unreactive metals do not react with dilute acids, so no hydrogen is given off.